2025/09 PHY301 Thermodynamics and Statistical Physics

Problems set 1

Lecturer: Song Kok Wee, kokwee.song@xmu.edu.my

Problem 1: Liouville's theorem and entropy [25 marks]

I. In the lecture, we learn that the evolution of the microstates in small time δt is

$$q'_\alpha = q'_\alpha(\{p_\alpha\},\{q_\alpha\}) = q_\alpha + \dot{q}_\alpha(\{p_\alpha\},\{q_\alpha\})\delta t$$
$$p'_\alpha = p'_\alpha(\{p_\alpha\},\{q_\alpha\}) = p_\alpha + \dot{p}_\alpha(\{p_\alpha\},\{q_\alpha\})\delta t.$$

where $\{p_\alpha\} \equiv (p_1,\ldots,p_{3N})$ and $(q_\alpha) \equiv \{q_1,\ldots,q_{3N}\}$. Using the change of variable method and Hamiltonian mechanics to prove that the phase space volume is unchanged up to $\mathcal{O}(\delta t^2)$. Namely, $dp'_\alpha dq'_\alpha = dp_\alpha dq_\alpha$ where the Jacobian $\frac{\partial(q'_1,\ldots,q'_{3N},p'_1,\ldots,p'_{3N})}{\partial(q_1,\ldots,q_{3N},p_1,\ldots,p_{3N})} = 1 + \sum_{\alpha=1}^{3N}(\frac{\partial\dot{q}_\alpha}{\partial q_\alpha} + \frac{\partial\dot{p}_\alpha}{\partial p_\alpha})\delta t + \mathcal{O}(\delta t^2)$.[*Note.* You can argue how the δt^2 arise in the determinant and do not need to evaluate the it exactly.] **[10]**

II. The normalized ensemble density is a probability in the phase space Γ. This probability has an associated entropy $S(t) = \int d\Gamma \rho(\Gamma,t)\ln\rho(\Gamma,t)$

(a) Using the method of Lagrange multipliers, find the function $\rho_{\max}(\Gamma)$ that maximizes the functional $S[\rho]$, subject to the constraint of fixed energy, $\langle\mathcal{H}\rangle = \int d\Gamma\rho\mathcal{H} = E$. **[10]**

(b) Show that the solution to part (b) is stationary, that is, $\frac{\partial\rho_{\max}}{\partial t} = 0$. **[5]**

Problem 2: Maxwell-Boltzmann distribution [25 marks]

Consider the energy of individual atoms in a box with length L in each dimension be $\varepsilon = \frac{1}{2}m(v_x^2+v_y^2+v_z^2)+mgz$, where m is the mass, g is the gravity and z is the height ($z \in [0,L]0$, one may set $L\to\infty$ for simplicity).

(a) Using Maxwell-Boltzmann theory and letting $\mu = (v_x,v_y,v_z,x,y,z)$ be the states of the particle, at thermal equilibrium with temperature T, what is the probability distribution (ρ) for finding a particle in state μ? **[7]**

(b) Prove that the density (n) of the gas at equilibrium $n(z) \propto \exp\left[-\frac{mgz}{k_BT}\right]$. **[5]**

(c) What is the average energy of a particle $\bar{\varepsilon}$? **[5]**

(d) What is the energy fluctuation $\bar{\epsilon^2} - \bar{\epsilon}^2$? **[8]**

Problem 3: Relativistic ideal gas [25 marks]

N *indistinguishable* non-interacting relativistic particles move in *one* dimension subject to a

Hamiltonian

$$\mathcal{H}(\{p_i, q_i\}) = \sum_{i=1}^{N} [c|p_i| + U(q_i)], \quad \text{with } U(q_i) = \begin{cases} 0, & \text{for } 0 \le q_i \le L \\ \infty, & \text{otherwise} \end{cases}$$

Consider a microcanonical ensemble of total energy E.

(a) Given that

$$\int_{\sum_{i=1}^{N} c|p_i| \le E} d^N p_i = \frac{2^N (E/c)^N}{N!},$$

and using the result in (a), what is the available phase space volume?

(b) Compute the entropy $S(E, L, N)$.

(c) Calculate the one-dimensional pressure P.(*Hint* compare the first law of thermodynamics $TdS = dE + PdL - \mu dN$ with the differential form of entropy $dS(E, L, N)$. The chemical potential μ is a energy cost for adding and additional particle to the system. The work done by doing that is μdN.)

(d) Obtain the heat capacities C_L with constant volume and C_P with constant pressure.(*Hint.* For C_p, you have to account for the work done to the external pressure.)

(e) What is the probability $p(p_1)$ of finding a paritcle with momentum p_1?

Problem 4: Equidistant energy levels system in microcanonical ensemble [25 marks]

In the lecture, we discuss two-level system. Here, we consider N independent quantum oscillators with infinite number of energy level. The Hamiltonian is

$$\mathcal{H}(\{n_i\}) = \sum_{i=1}^{N} \hbar\omega \left(n_i + \frac{1}{2} \right),$$

where $n_i = 0, 1, 2, \ldots$ is energy level for the ith oscillator. In this system, the energy difference between adjacent levels is a constant $\hbar\omega$ [Stirling approximation $\ln N! \approx N \ln N - N$]

(a) Using microcanonical ensemble, calculate the entropy S, as a function of total energy E. [*Hint.* The total number of microstates $\Omega(E)$ can be regarded as the number of ways by permuting $M = \sum_i n_i$ balls and $N - 1$ partition walls together along a line.]

(b) Calculate the energy E, and the heat capacity C, as functions of temperature T, and N.

(c) Find the probability $p(n)$ that a particular oscillator is in its nth quantum level.